

J.P. MORGAN PAYMENT CARTRIDGE FOR SALESFORCE COMMERCE CLOUD (SFRA)

Implementation Guide

Document Control

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Supported Cards	Direct API - Visa, Mastercard, American Express, Discover Drop-in – Decided during merchant On Boarding
Supported Currency	USD, CAD, EUR, GBP
Supported Payment Methods	Credit Card, Google Pay, Apple Pay
Supported Locale <i>(Supports multi-locale)</i>	North America: en_US, en_CA



	• Europe: Supports all EU regional locales.
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Revision History

Version	Region	Supported Locales	Integration Type
v1.0	North America	en_US, en_CA	Standard Integration
v2.0	Europe (EU)	Standard EU Regional Locales	Drop-in Integration

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1. PURPOSE AND SCOPE

1.1 Purpose

This guide provides step-by-step instructions to install, configure, test, and deploy the J.P. Morgan Payment Cartridge for Salesforce Commerce Cloud (SFCC) using the Storefront Reference Architecture (SFRA).

The cartridge implements a Dual-Integration Architecture, supporting both Direct API (utilizing Page Integrated Encryption) and Drop-in UI functionality.

1.2 Scope

In scope for Direct API (US & CA):

- *Card Payment Processing: Seamless credit card processing directly in the storefront.*
- *Data Security: Page Integrated Encryption (PIE) and SafeTech tokenization.*
- *Account Updater: Real-time account updates handled natively.*
- *Fraud Check: Optional configuration during pre-authorization.*
- *Business Manager Operations: Customer Service Center (CSC) portal actions including capture, refund, and void.*
- *Multi-Merchant Support: Individual Merchant ID (MID) configurations mapped per locale.*
- *Digital Wallets: Express Checkout and standard checkout payments via Google Pay and Apple Pay.*
- *Deployment: Comprehensive testing, validation, and go-live readiness steps.*

In scope for Drop-in API (EU):

- *Card Payment Processing: Hosted drop-in payment interface rendering during checkout.*
- *Profile Tokenization: Ability to save cards to a consumer profile for subsequent checkout sessions.*
- *Account/Order Updater: Asynchronous backend updates via platform notifications.*
- *Business Manager Operations: Customer Service Center (CSC) portal actions including capture, refund, and void.*
- *Multi-Merchant Support: Individual Merchant ID (MID) configurations mapped per locale.*
- *Digital Wallets: Standard checkout payments via Google Pay and Apple Pay.*
- *Deployment: Comprehensive testing, validation, and go-live readiness steps.*

***Remember that SafeTech Tokenization applies to Direct API, whereas Drop-in uses JPMC-hosted profile saving.*

2. OVERVIEW

The J.P. Morgan Payment Cartridge integrates J.P. Morgan's payment processing services with an SFCC SFRA storefront using Direct API integration and Drop-in Integration.

2.1 Key Capabilities

Direct API Capabilities (US & CA Only)

- *Page Integrated Encryption (PIE): Encrypts card data directly in the browser before it reaches the server, significantly reducing your PCI DSS compliance scope.*
- *SafeTech Tokenization: Replaces Primary Account Numbers (PANs) with secure tokens for safe card storage and subsequent payments.*
- *Real-Time Account Updater: Automatically refreshes expired or replaced credit cards in real-time within the customer's "My Account" section.*
- *Express Checkout: Optionally displays Google Pay and Apple Pay buttons on Product Detail Pages (PDP) and the Cart page, allowing customers to bypass the standard checkout flow.*
- *Optional Fraud Check: Provides pre-authorization risk scoring to automatically accept, review, or block suspicious transactions.*

Drop-in UI Capabilities (EU Only)

- *Secure Drop-in UI: Renders a secure, J.P. Morgan-hosted payment interface during checkout, streamlining local European payment processing and compliance requirements.*

- *Asynchronous Account & Order Updates: Automatically refreshes expired or replaced credit cards and syncs order status updates asynchronously via platform notifications.*

Shared Capabilities (Direct API & Drop-in UI)

- *Major Card Acceptance: Securely accept all major credit cards (Visa, Mastercard, American Express, and Discover) processed through J.P. Morgan's secure infrastructure.*
- *Digital Wallets: Support seamless payments via Google Pay and Apple Pay during the standard checkout flow.*
- *Business Manager Extensions: Perform key operational actions—such as capture, refund, and void—directly from the Customer Service Center (CSC) order view in Salesforce.*
- *Multi-Merchant Support: Configure individual Merchant IDs (MIDs) per locale, all managed from a single Salesforce Commerce Cloud site.*

2.2 Integration Models: Direct API (PIE) vs. Drop-in

The J.P. Morgan Payment Cartridge supports two distinct integration architectures at the storefront level. The active integration model is determined dynamically at the site level via Site Preferences:

- *Direct API (PIE): Designed for North American markets (US and CA). This model utilizes Page Integrated Encryption (PIE) to secure cardholder data directly from the browser, giving the merchant total control over the checkout UI.*
- *Drop-in: Designed for European markets (EU). This model injects a secure, pre-built J.P. Morgan checkout interface onto your payment page, simplifying local compliance and automatically rendering regional European payment methods.*

2.3 Regional Configuration Matrix

To ensure successful transaction routing, merchants must configure site preferences strictly according to regional compatibility.

Storefront Region	Required Integration Model
United States (US)	Direct API (PIE)
Canada (CA)	Direct API (PIE)
Europe (EU Countries)	Drop-in

⚠ ****CRITICAL CONFIGURATION RULE:**** Direct API (PIE) is NOT supported for EU storefront configurations.

Attempting to cross-configure these models (e.g., setting an EU site to PIE) will result in checkout failures.

2.4 Process Flow for Direct API (High Level)

2.4.1 Process Flow Credit Card (High Level)

1. Shopper enters card details at checkout.
2. PIE scripts encrypt card data in the shopper's browser.
3. Cartridge submits encrypted payment payload for authorization.
4. J.P. Morgan returns authorization outcomes and SafeTech token.
5. Capture occurs immediately or later depending on configured capture method.

****Why does this matter? **** - Because the real card number is encrypted before it even leaves the customer's browser. This reduces the security burden and helps with PCI compliance.

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2.4.2 Process Flow Google Pay (High Level)

- Shopper taps Google Pay button
- Sends payment request to Google Pay
- Google shows payment details. Customer uses their saved card in Google account
- Payment request is sent using encrypted payment data received from Google to J.P. Morgan
- Capture occurs immediately or later depending on configured capture method.

2.4.3 *Process Flow Apple Pay (High Level)*

- *Shopper taps Apple Pay button*
- Sends payment request to Apple Pay
- Apple shows payment details. Shopper approves with Face ID or Touch ID.
- Payment request is sent using encrypted payment data received from Apple to J.P. Morgan
- Capture occurs immediately or later depending on configured capture method.

2.5 Process Flow for Dropin (High Level)

- Shopper lands on checkout: The shopper navigates to the billing page, prompting the secure, JPMC-hosted Drop-in UI to load automatically.
- Shopper selects and pays: The shopper chooses their preferred payment method, enters any required details directly within the hosted interface, and clicks "Pay Now."

- Order confirmation: The cartridge processes the secure payment response, places the order in Salesforce, and directs the shopper to the order confirmation page.

3. FEATURE COMPARISION BY INTEGRATION MODEL

The features available to your storefront customers are strictly tied to the integration model configured for that specific site. Use this matrix to understand the regional feature differences before designing your customer journeys:

Feature	Direct API (PIE)	Drop-In
	US/ CA	EU
Supported Credit Cards	Yes	Yes
Apple Pay on Checkout Page	Yes	Yes
Google Pay on Checkout Page	Yes	Yes
Apple Pay / Google Pay on PDP & Cart	Yes	No (Unsupported)
Saved Cards in 'My Account'	Yes	No (Unsupported)
Account Updater (Real-time in My Account)	Yes	No
Account Updater (Checkout via Notifications)	No	Yes

4. SOLUTION COMPONENTS

Think of this integration like building blocks that work together. Here is what each piece does:

*****What is a cartridge? *****

In Salesforce Commerce Cloud, a "cartridge" is like an app or plugin. You install it on your site to add new features.

Cartridge	Purpose
`int_jpmc_sfra`	SFRA storefront overlay (checkout UI, hooks, client-side scripts)
`int_jpmc_core`	Core payment logic (services, authentication, helpers)
`bm_jpmc`	Business Manager extensions (CSC payment actions, tools)

SFCC Business Manager — The Control Panel

*****Business Manager***** is the admin area of your Salesforce Commerce Cloud site. Think of it as the "back office" where you manage products, orders, and settings.

4. PREREQUISITES AND REQUIREMENTS

4.1 Inputs Required from J.P. Morgan Payments Representative

Shared Core Credentials (Both Direct API & Drop-in UI)

- Merchant ID (MID): Your unique J.P. Morgan merchant identifier used to route and settle transactions.
- Client ID: The OAuth2 client identifier.

Direct API Configuration (US & CA Only)

Core Credentials

- PIE Key(Needed for Direct API only): The Page Integrated Encryption (PE) Key ID used for secure frontend data transmission.

Digital Wallets & Fraud

- Google Pay Gateway Merchant ID: Required exclusively for Production environments to route Google Pay transactions.
- Apple Pay MID: Apple Pay Merchant ID created in Apple Developer portal.
- Apple Pay Payment Processing Certificate: A mandatory prerequisite for Apple Pay integration. This certificate must be obtained and uploaded directly to your Apple Developer Portal.
- SafeTech Client ID: Required only if you are utilizing the integrated SafeTech fraud detection capabilities.

** Either Merchant can generate a CSR via OpenSSL and share with J.P Morgan to receive a signed certificate or share a signed certificate

obtained from one of the certificate authorities listed in the J.P Morgan documentation.

Documentation Reference:

For detailed certificate specifications and clarifications, refer to

[oauth-authentication](#)

Check with J.P Morgan for any clarifications on the certificate.

4.2 SFCC Environment Requirements

Ensure your Salesforce Commerce Cloud (SFCC) platform and local development environments meet the following prerequisites before proceeding:

Platform Prerequisites

- SFCC Instance: Access to a working SFCC environment (Sandbox, Development, or Staging).
- Business Manager Access: Right Privileges to configure Site Preferences, Services, Custom Objects, Customer Service Centre Settings, Business Manager Modules Settings, and Certificates.

5. INSTALLATION

****Technical Notice****: The following steps involve uploading files to your Salesforce BM and require using a ****command-line tool**** (Terminal on Mac or Command Prompt on Windows).

5.1 Before you install the JPMC cartridge, make sure that you:

- *Have access to the Salesforce Sandbox.*
- *Installed the [Storefront Reference Architecture \(SFRA\) repository on GitHub](#). You need developer access to access the repository on GitHub.*
- *Downloaded the JPMC cartridge from our [GitHub repository](#).
[TODO: THIS NEEDS TO BE CHANGED TO RIGHT URL]*

Local Development Prerequisites

- Node.js: Version 24.x or higher is recommended for local compilation and script tasks.
- Package Manager: npm (included with Node.js) installed on your local workstation to manage project dependencies.

5.2 Create and Configure `dw.json` for Cartridge upload to the Business Manager

**** Goal****: Create a small configuration file that tells your computer where your Salesforce site is located so the upload tool knows where to send the cartridges.

Configure sandbox credentials in the root of the `link\_jpmc` folder.

> See Appendix A for a standardized sample.

5.3 Install Dependencies in link_jpmc root folder

****Goal**:** Download all the helper tools the cartridge needs.

1. Open a Terminal or Command Prompt
2. Navigate to the `link\_jpmc/` folder:

```
```bash
 cd path/to/link_jpmc
```
```

3. Run:

```
```bash
 npm install
```
```

4. Wait for the installation to complete.

5.3 Build Client-Side Assets in link_jpmc root folder

****Goal**:** Compile the JavaScript and CSS files that your storefront needs.

Run:

```
```bash
 npm run build
```
```

5.4 Upload Cartridges

****Goal****: Send the cartridge files from your computer to your Salesforce sandbox.

Upload the following cartridges (WebDAV / Prophet / equivalent):

- `int\_jpmc\_sfra`
- `int\_jpmc\_core`
- `bm\_jpmc`

Next step: Proceed to 6. Metadata Import.

6. METADATA IMPORT

6.1 Purpose

Metadata import creates required services, site preferences, custom objects, and object extensions in Business Manager.

**** This section involves importing XML files into your Salesforce site.**
If you are not comfortable with file imports, please ask your web developer or IT support person for help.

****Goal**:** Tell the SFCC BM, about all the new data fields, services, jobs, payment-methods, custom processor and custom objects the J.P. Morgan cartridge uses.

| File | What It Does |
|-----------------------------------|---|
| system-objecttype-extensions.xml | Creates all the JPMC configuration fields in Business Manager (like "JPMC Client ID," "Capture Method," etc.) and adds payment data fields to orders. |
| custom-objecttype-definitions.xml | Creates the behind-the-scenes data storage objects — access tokens, merchant configurations. |
| services.xml | Sets up the connection endpoints (URLs) that your store uses to communicate with J.P. Morgan's servers. |
| payment-methods.xml | Defines the available payment options (e.g., Credit Card, Google Pay Payments) visible at checkout and maps them to the appropriate custom processor. |

| | |
|------------------------|--|
| payment-processors.xml | Registers the custom payment processors and maps them to the specific cartridge scripts responsible for handling payment hooks and transaction states. |
| jobs.xml | Configures the Job Framework, defining the background tasks and schedules used for automated operations. |

6.2 Prepare the Import Archive

To prepare the JPMC integration metadata for your specific environment, follow the steps below within the `link_jpmc/metadata` directory of the codebase.

Step 1: Configure the Site Directory

- *Navigate to the following path: `link_jpmc/metadata/site_import/sites/`.*
- *Locate the folder named `RefArch` (the default reference architecture ID).*
- *Rename the `RefArch` folder to match your Target Site ID exactly as it is configured in Business Manager (e.g., `YourBrandID`).*

Step 2: Update Job Metadata

- *Navigate to `link_jpmc/metadata/site_import/` and open the `jobs.xml` file.*
- *Locate any instances of the `site-id` attribute or element.*
- *Replace the placeholder value with your specific Target Site ID. This ensures that the scheduled jobs are correctly mapped to your storefront during the import process.*

Step 3: Package for Site Import

- *Return to the `link_jpmc/metadata/` directory.*

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- Compress the `site_import` folder into a `.zip` archive.
****Important****: Ensure the archive structure is correct for SFCC Site Import. The root of the zip should contain the `sites/`, `jobs.xml`, `meta`, `services` components.

For a reference on the required structure, you can inspect the existing sample at: `link_jpmc/metadata/site_import.zip`.

6.3 Upload and Import in Business Manager

- Administration > Site Development > Site Import & Export
- Under Import, click Upload and select ``site_import.zip.zip``.
- Select the uploaded zip and click Import.
- Confirm import completes with SUCCESS.

[Administration](#) > [Site Development](#) > Site Import & Export

Site Import & Export

This page allows you to export the current configuration of your organization including all of its sites. To download an archive, just click its file name.

Import

Upload Archive:

☒ Local ☐ Remote

No file chosen

6.4 Objects and Configurations Created

| File | Creates |
|-----------------------------|---|
| <code>`services.xml`</code> | HTTP services, credentials, service profile |

| | |
|-------------------------------------|--|
| `system-objecttype-extensions.xml` | Site preferences and groups, Order and Payment custom attributes |
| `custom-objecttype-definitions.xml` | `JPMCAccessToken` custom object for OAuth token + merchant configurations + Notifications Queue |
| payment-methods.xml | JPMC Payment methods |
| payment-processors.xml | Custom processor JPMC_Payment |
| jobs.xml | Scheduled Jobs scheduled jobs, steps, and flows responsible for automated background processes within the integration. |

7. CARTRIDGE SETUP (BM CONFIGURATION)

7.1 Update Storefront Cartridge Path

- *Administration > Sites > Manage Sites*
- *Select your storefront site.*
- *Settings tab > Cartridges*
- *Add `int\_jpmc\_sfra` and `int\_jpmc\_core` before `app\_storefront\_base`:*

```
`int_jpmc_sfra:int_jpmc_core:app_storefront_base`
```

- *Click Apply.*

[Administration](#) > [Sites](#) > [Manage Sites](#) > RefArch - Settings

General **Settings** Cache Site Status Page Meta Tag Rules

RefArch - Settings

Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

Instance Type:

Deprecated. The preferred way of configuring HTTP and HTTPS hostnames is by using new features of the site aliases configuration ("SEO > Aliases Configuration"). The HTTP/HTTPS hostname values set in t

HTTP Hostname:

HTTPS Hostname:

Instance Type: All

Cartridges:

7.2 Update Business Manager Cartridge Path

- *Administration > Sites > Manage Sites*
- *Select Business Manager.*
- *Settings tab > Cartridges*
- *Add `bm\_jpmc` and `int\_jpmc\_core` before `bm\_app\_storefront\_base`:*

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``bm_jpmc:int_jpmc_core:bm_app_storefront_base``

- 5. Click Apply.

[Administration](#) > [Sites](#) > [Manage Sites](#) > Business Manager - Settings

[Settings](#)
[Cache](#)
[Hostnames](#)

Business Manager - Settings

Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

Instance Type: Sandbox/Development

Deprecated. Up to two instance specific hostname aliases for Business Manager can be configured here.
 ⚠ Setting a different hostname here will make some Business Manager modules unreachable. Manage additional hostnames in the 'Hostnames' tab instead.

HTTP Hostname:

HTTPS Hostname:

Instance Type: All

Cartridges: bm_jpmc:int_jpmc_core:bm_app_storefront_base

7.3 Verify Services are Available

1. Administration > Operations > Services
2. Confirm these services appear

Core Service (For both Direct API and Drop-in)

1. Authentication & Security
 - Service: JPMCAccessToken
 - Purpose: Handles OAuth 2.0 authentication and bearer token generation.
 - Credential ID: JPMCAccessTokenCredentials

Direct API Services

1. Core Payment Lifecycle
 - Service: JPMCPaymentService
 - Purpose: Processes initial payment authorizations.
 - Credential ID: JPMCPaymentServiceCredentials

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- Service: JPMCPaymentCapture
 - Purpose: Executes capture requests for authorized payments.
 - Credential ID: JPMCPaymentCaptureCredentials
- Service: JPMCPaymentRefund
 - Purpose: Manages transaction refunds for processed orders.
 - Credential ID: JPMCPaymentRefundCredentials
- Service: JPMCPaymentVoid
 - Purpose: Voids authorized payments prior to capture.
 - Credential ID: JPMCPaymentVoidCredentials

2. Risk & Verification

- Service: JPMCFraudCheck
 - Purpose: Interfaces with JPMC fraud tools to assess transaction risk.
 - Credential ID: JPMCFraudCheckCredentials
- Service: JPMCVerification
 - Purpose: Performs account validations (e.g., zero-dollar auth).
 - Credential ID: JPMCVerificationCredentials

Checkout & Drop-in Services (Drop-in UI Only)

Checkout:

- *Service: JPMCCheckoutIntent*

- Purpose: Generates a secure checkout intent session with J.P. Morgan, establishing the transaction payload required to securely initialize and render the hosted Drop-in UI during checkout.
- Credential ID: JPMCCheckoutIntentCredentials

Notifications :

- Service: JPMCNotificationsReceive
 - Purpose: Listens for and receives platform notifications from J.P. Morgan regarding order.
 - Credential ID: JPMCNotificationsReceiveCredentials
- Service: JPMCNotificationsAck
 - Purpose: Acknowledges receipt of J.P. Morgan platform notifications to successfully close the notification loop and prevent redundant retries.
 - Credential ID: JPMCNotificationsAckCredentials

⚠ Important: Ensure CAT or Production URLs are configured in each credential. Current configurations have Mock URLs.

8. CERTIFICATES AND KEYS

To ensure the highest level of security, all communication between site and J.P. Morgan is ****digitally signed**** with a certificate. The certificate proves to J.P. Morgan that payment requests are really coming from you and have not been tampered with.

8.1 Purpose

The cartridge uses an X.509 certificate and private key to sign JWT tokens for API authentication.

J.P Morgan uses the certificate thumbprint (SHA-1) from the access token generated and verifies the authenticity of the merchant.

Note: Refer documentation for acceptable formats for Certificate and Private Keys:

[Import-3rd-party-private-keys-and-certificates-for-an-instance](#)

8.2 Upload Certificate and Private Key

1. Administration > Operations > Private Keys and Certificates
2. Click Import Key/Certificate
3. Upload private key with alias: `jpmc-private-key`. Here you are supposed to enter the password that was used during key generation.
4. Upload certificate with alias: `jpmc-certificate`

Important: Alias values must match what you configure in site preferences/custom Object (Multi-Merchant Support).

Private Keys and Certificates ⓘ

| ☐ Alias | Hostnames | Type |
|---|-----------|---------------------|
| ☐ jpmc-certificate | | Trusted Certificate |
| ☐ jpmc-private-key | | Private Key |

9. KEY ID (KID) GENERATION – ONE TIME ACTIVITY

What Is a Key ID?

After you upload the certificate, you need to generate its unique "fingerprint" — called a ****Thumbprint**** or ****Key ID (kid)****. This fingerprint proves to J.P. Morgan that requests from your store were signed with your specific certificate.

9.1 Purpose

The Key ID (`kid`) is a SHA-1 thumbprint of your certificate used in the JWT header.

9.2 Generate Thumbprint in Business Manager

To See **JPMC Tools > Certificate Thumbprint Generator** in Merchant Tools

- **Administration > Organization > Roles & Permissions**
 > *SELECT A ROLE (Eg: Administrator) > Business Manager Modules*
- *Select Site (context)*
- *Click Certificate Thumbprint Generator*
- *Click Update*

[Administration](#) > [Organization](#) > [Roles](#) > Administrator - Business Manager Modules

General Users **Business Manager Modules** Functional Permissions WebDAV Permissions Locale Permissions Price Adjustment Limits Customer Service Center Permissions

Administrator - Business Manager Modules

This list shows all Business Manager modules available in the system for which permissions can be granted. Click **Select Context** to select the context for which you would like to modify permissions. Available context options are the organization, one site, or multiple sites. Select the checkboxes and use the **Update** button at the bottom of the page to grant permissions to certain Business Manager modules. Deselect the checkboxes and use the **Update** button to revoke permissions to specific Business Manager modules.

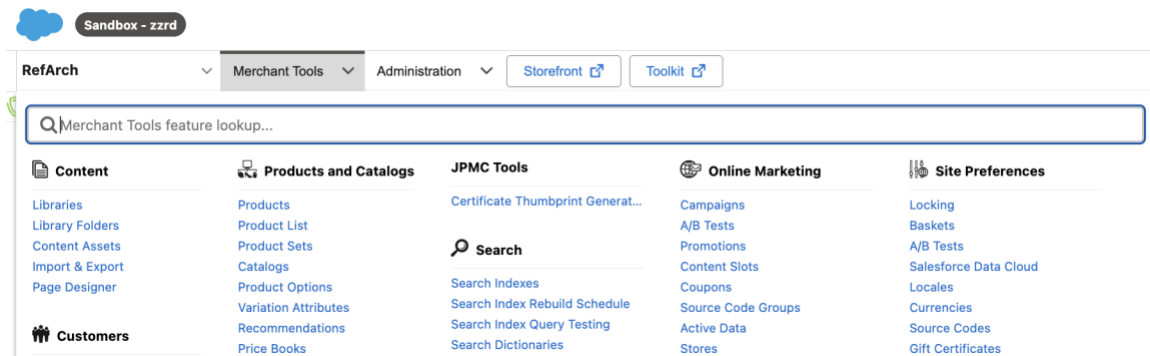
Read access may be granted to an increasing number of Business Manager modules. Write access includes read access. When removing read access, make sure you haven't also granted write access for the respective feature.

When multiple contexts are selected with different permissions for a feature or module, detailed information can be viewed by clicking in the "Details" column.

| Selected Context: RefArch | | | |
|----------------------------------|---|-------------------------------------|-------------------------------------|
| Select Context | | | |
| Business Manager Module | Module Description | Read | Write |
| JPMC Tools | | ☐ | ☐ |
| Certificate Thumbprint Generator | Generate SHA-1 thumbprint from BM Certificate Manager | ☒ | ☒ |

9.3 Generate Thumbprint in Business Manager

- *Merchant Tools > JPMC Tools > Certificate Thumbprint Generator*
- *Enter Configuration Scope (Where do you want to save data)*
- *Enter Certificate Alias (default: `jpmc-certificate`)*
- *Enter Private Key Alias (default: `jpmc-private-key`)*
- *Click Generate & Save*



10. ORGANIZATION PREFERENCE CONFIGURATION

10.1 Open Preference Groups

Configure the mandatory Realm ID (JPMC) under

Administration > Global Preferences > Global Custom Preferences > **JPMC Global Configurations > JPMC Realm ID**

[Administration](#) > [Global Preferences](#) > [Global Custom Preferences](#) > JPMC Global Configuration

Global Custom Preferences

This page allows you to define the global custom preferences of group 'JPMC Global Configuration'. For each instance type,)

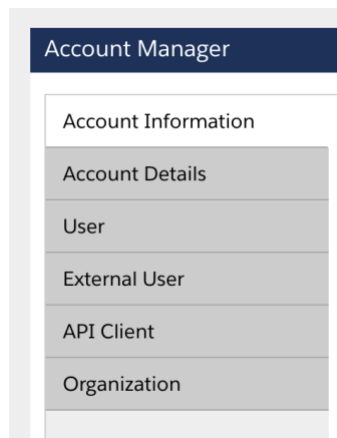
| Instance Type: <input type="text" value="Sandbox/Development"/> | |
|---|----------------------|
| Preference Name | Value |
| SFCC Instance Realm ID:* | <input type="text"/> |

How to obtain Realm ID?

1. Login to your Account Manager.

<https://account.demandware.com/dw/oidc/authorization/openAm>

2. Click on Organization on left pane.



3. Click on your Organization from the list.



The screenshot shows the 'Account Manager' interface. On the left is a sidebar with a menu containing 'Account Information', 'Account Details', 'User', 'External User', 'API Client', and 'Organization'. The 'Organization' item is selected. The main content area is titled 'Organizations' and contains the text 'Edit your organizations.' Below this is a table with one row. The table has a header 'Organization Name ▲▼' and a single entry 'JPMorgan Chase & Co' with a blue link icon to its right.

| Organization Name ▲▼ |
|----------------------|
| JPMorgan Chase & Co |

4. Scroll down to Assigned Realms section.

Assigned Realms

| ID | Description |
|---------------|-------------|
| z[REDACTED]ix | |

5. Copy the 4 lettered Realm ID associated to the organization.

11. SITE PREFERENCES CONFIGURATION

11.1 Open Preference Groups

Configure the J.P. Morgan Commerce (JPMC) payment integration through Merchant Tools > Site Preferences > Custom Preferences.

This section outlines all configuration groups.

1. **JPMC Core Settings:** Foundational credentials, authentication, and core operational toggles.
2. **JPMC Drop-in Settings:** Configurations specifically for the hosted Drop-in UI.
3. **JPMC Direct Payment API Settings:** Configurations for PIE encryption, tokenization, Account Updater, Google Pay, Fraud, and 3DS.
4. **JPMC PWA Configuration:** Settings only applicable for Progressive Web App (PWA) integrations, including Apple Pay.

11.2 Configuration Groups

11.2.1 JPMC Core Settings

Navigate to: Merchant Tools > Site Preferences > Custom Preferences > JPMC Core Settings

1. **Integration Type (JPMCCheckoutMode):** Toggles between the standard Page-Integrated Encryption card form (PIE) and the European hosted checkout UI (DROP_IN).
2. **JPMC Client ID (JPMCClientID):** OAuth2 Client ID from JPMorgan. (String/Password)

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3. **JPMC Merchant Id (JPMC_MerchantCode):** Merchant ID from JPMorgan (e.g., *YOUR_merchantId*). (String)
4. **JPMC Merchant Category Code (JPMCMerchantCategoryCode):** Merchant Category Code from JPMorgan (4 Digited number).
5. **JPMC Resource ID (JPMCResourceID):** OAuth2 resource ID. (Enum: Test/CAT, Production). Default: TEST.
6. **JPMC Certificate Alias (JPMCCertAlias):** Alias of the X.509 certificate in BM Certificate Manager.
7. **JPMC Private Key Alias (JPMCPrivateKeyAlias):** Alias of the private key in BM Certificate Manager.
8. **JPMC Key ID (jpmc_kid):** Key ID used for authenticating payloads.
9. **JPMC Capture Method (JPMCCaptureMethod):** Sets when funds are captured: MANUAL (authorize only), DELAYED (delayed auto-capture within a 120-minute window), or NOW (immediate sale). Default: MANUAL.
10. **Enable Multi-Merchant Configuration (JPMCEnableMultiMerchant):**
 - DISABLED (False): Entire site uses single merchant config from Site Preferences.
 - ENABLED (True): Uses per-locale/per-merchant configurations from custom objects (JPMCMerchantConfig) and cache. Site Preferences act as a fallback. (Boolean)

11.2.2 JPMC Drop-in Settings

Navigate to: Merchant Tools > Site Preferences > Custom Preferences > JPMC Drop-in Settings

1. **Drop-in UI Script URL (JPMCDropInScriptUrl):** The hosted JavaScript module URL used to render the Drop-in UI (configured for either Test/CAT or Production).
2. **Drop-in UI Theme Overrides (JPMCDropInThemeOverrides):** An optional JSON object used to customize the Drop-in UI's appearance (colors, borders, fonts) to match your storefront. Leave empty to use Commerce Center defaults.
3. **Save Consumer Profile (JPMCSaveConsumerProfile):** When enabled (true), the Drop-in checkout session is created so J.P. Morgan securely saves and manages the customer's payment profile on their servers for future checkouts. (Boolean)
4. **Last Sync Period End (JPMC_LastSyncPeriodEnd):** Tracks the timestamp of the last successful synchronization period.

11.2.3 JPMC Direct Payment API Settings

Navigate to: Merchant Tools > Site Preferences > Custom Preferences > JPMC Direct Payment API Settings

URLs & Encryption (PIE)

1. **JPMC PIE Get Key Base URL (JPMCGetKeyUrl):** Base URL for JPMC PIE getkey.js script. Default: TEST.
2. **JPMC PIE Encryption Library URL (JPMCEncryptionUrl):** URL for JPMC Page Integrated Encryption (PIE) script. Default: TEST.
3. **JPMC PIE Key: Page Encryption Key ID (JPMCPieKey):** PE ID - Appended to construct the full URL. (String)

Tokenization & Account Updater

1. **JPMC Tokenization Type (JPMCTokenizationType):** Default tokenization type for transactions (Enum: SAFETECH_TOKEN, NETWORK_TOKEN). Default: SAFETECH_TOKEN.

2. **Account Updater Mode (jpmcAccountUpdaterMode):** Controls active flows: NONE, NOTIFICATIONS (async webhook), REAL_TIME (RTAU on auth), BOTH.

Google Pay Configuration

1. **Google Pay Environment (JPMCGooglePayEnvironment):** TEST or PRODUCTION. Default: TEST. Choose Test while setting up. Switch to Production only when ready for real customer payments.
2. **Google Pay Gateway Merchant ID (JPMCGooglePayGatewayMerchantId):** The merchant ID that J.P. Morgan provided to you for Google Pay.
3. **Google Pay Merchant ID (JPMCGooglePayMerchantId):** Your Merchant ID from the Google Pay Business Console. Leave blank for Test mode. Required for Production.
4. **Google Pay Merchant Name (JPMCGooglePayMerchantName):** The name your customers see in the Google Pay pop-up.
5. **Google Pay Gateway (JPMCGooglePayGateway):** The payment gateway name, typically chase.
6. **Google Pay Allowed Card Networks (JPMCGooglePayAllowedCardNetworks):** A comma-separated list of card types you accept (e.g., VISA,MASTERCARD,AMEX,DISCOVER).
26. **Google Pay Allowed Auth Methods (JPMCGooglePayAllowedAuthMethods):** A comma-separated list of how cards can be authenticated (e.g., PAN_ONLY,CRYPTOGRAM_3DS).
7. **JPMC Google Pay Cart Enabled (JPMCGooglePayCartEnabled):** Toggles to enable Express Checkout buttons on the Cart. (Boolean) Default: NO.
8. **JPMC Google Pay PDP Enabled (JPMCGooglePayPDPEntered):** Toggles to enable Express Checkout buttons on Product Detail Pages. (Boolean) Default: NO.

Platform & Fraud Verification

1. **JPMC Platform ID (JPMCPlatformId):** Optional platform identifier. (String)
2. **Enable JPMC Fraud Check - Handle (JPMCEnableFraudCheck):** Validates fraud during payment verification (basket stage). Declines block the transaction. (Boolean) Default: NO.
3. **Enable JPMC Fraud Check - Authorize (JPMCEnableFraudCheckAtAuth):** Validates fraud before payment authorization (order stage). (Boolean) Default: NO.
4. **Safetech Fraud Client ID (jpmcKountClientId):** Safetech Fraud Web Client SDK Client ID (6-digit ID). Required if fraud check is enabled. (String)
5. **Safetech Fraud Environment (jpmcKountEnvironment):** Enum: TEST or PROD. Default: TEST.
6. **Enable Address Verification Service (JPMCEnableAVS):** Includes billing address in Verify Payment and Authorization payloads. (Boolean) Default: NO.

3D Secure

1. **Enable JPMC 3D Secure (jpmc3DSEnabled):** * DISABLED (False): Disables 3DS Feature. * ENABLED (True): Enables 3DS Feature. (Boolean)

11.2.4 JPMC PWA Configuration

Navigate to: Merchant Tools > Site Preferences > Custom Preferences > JPMC PWA Configuration

(Note: These settings are exclusively applicable for headless/PWA integrations)

1. **Apple Pay Enabled (JPMCApplePayEnabled):** Toggles to enable or disable Apple Pay as an active payment method in the PWA storefront. (Boolean) Default: NO.

2. **Apple Pay Merchant ID (JPMCAApplePayMerchantId):** The unique Apple Pay Merchant Identifier registered and configured in your Apple Developer portal. (String)
3. **Apple Pay Merchant Name (JPMCAApplePayMerchantName):** The display name shown to the customer on the Apple Pay payment sheet (typically your storefront brand name). (String)
4. **Apple Pay Country Code (JPMCAApplePayCountryCode):** The two-letter ISO country code representing the location of the processing merchant (e.g., US or GB). (String)
5. **Apple Pay Supported Networks (JPMCAApplePaySupportedNetworks):** A comma-separated list of the accepted credit/debit card networks for Apple Pay checkouts (e.g., visa, mastercard, amex, discover). (String)
6. **Apple Pay Merchant Capabilities (JPMCAApplePayMerchantCapabilities):** The security and processing capabilities supported by your Apple Merchant account, typically containing supports3DS. (String)
7. **API Host (JPMCApiHost):** The target base URL / host endpoint used by the PWA's backend services to send payment API requests to J.P. Morgan. (String)
8. **Environment (JPMCEEnvironment):** Defines the operational mode for headless PWA API connections (e.g., TEST or PRODUCTION). Default: TEST.
9. **Google Pay Allowed Shipping Countries (JPMCGooglePayAllowedShippingCountries):** A comma-separated list of two-letter ISO country codes representing valid shipping destinations supported during Google Pay Express checkouts. (String)
10. **Drop-in Intent URL (JPMCDropInIntentUrl):** The specific secure gateway endpoint URL used to retrieve or construct the

hosted Drop-in UI checkout session intents in a PWA architecture. (String)

12. MULTI-MERCHANT SUPPORT

12.1 What is Multi-Merchant Functionality?

By default, the cartridge uses a single JPMC merchant account for all transactions. The multi-Merchant feature allows you to override this and assign a unique set of JPMC gateway credentials to each locale (e.g., your Canadian English site, your US site, etc.).

This means each locale can have its own independent:

- *Merchant ID (MID)*
- *Fraud and AVS rules*
- *Currency processing rules*
- *Security certificates and keys*
- *Google Pay settings*

Example:

If a shopper uses this locale... | ...their payment is processed by this account:

en_CA (English Canada) | Canada (CAD) account

en_US (English US) | US (USD) account

it_IT (Italian Italy) | Italy (EUR) account

en_GB (English UK) | UK (GBP) account

12.2 When Do You Need It? A Checklist

You may use this feature if any of the following apply to your business:

- *You operate in multiple countries and have separate merchant accounts for each.*
- *You process payments in multiple currencies, and your acquirer requires a different MID per currency.*
- *You need to apply different fraud rules (e.g., AVS requirements) for specific regions.*
- *You need to use distinct Google Pay merchant identifiers for different countries.*
- *If you operate in a single country with a single currency, you do not need this feature.*

12.3 Setup Process

Part 1: The Setup Process (5 Steps)

Follow these five steps to configure a locale with its own merchant account.

Step 1: Gather Your Credentials

Before you begin, ensure you have the following information from JPMC for each locale you plan to configure:

- *Merchant ID (MID)*
- *Client ID*
- *PIE KEY*
- *Certificate Details: The names (aliases) of the certificate and private key you've uploaded to SFCC Business Manager.*

Step 2: Enable the Feature

First, you must activate the functionality for your site.

- Navigate to: **Merchant Tools** → **Site Preferences** → **Custom Preferences**.
- Select the JPMC Multi Merchant group.
- Find Enable Multi-Merchant and set it to Yes.
- Click Save.

This setting can be turned off at any time to revert to using a single, site-wide merchant account without losing your saved configurations.

Step 3: Create a New Locale Configuration

To See **JPMC Tools > Multi-Merchant Configuration in Merchant Tools**

- **Administration > Organization > Roles & Permissions > SELECT A ROLE (Eg: Administrator) > Business Manager Modules**
- Select Site (context)
- Click Multi-Merchant Configuration
- Click Update

| Business Manager Module | Module Description | ☐ | ☐ |
|--|---|--------------------------|-------------------------------------|
|  JPMC Tools | | | ☒ |
|  Certificate Thumbprint Generator | Generate SHA-1 thumbprint from BM Certificate Manager | | ☒ |
|  Multi-Merchant Configuration | Create and Manage per-locale JPMC merchant configurations | | ☒ |

Next, you will create the configuration for a specific locale.

- Navigate to: **JPMC Tools** → **Multi-Merchant Configuration**.
- Click the + Create / Edit Configuration button.
- From the Select Locale dropdown, choose the locale you want to configure (e.g., en_US, en_CA)

Merchant Configurations

About Merchant Configuration

This tool allows you to manage locale-specific payment gateway configurations for your JPMC integration. Each locale (e.g., en_US, en_CA) can have its own merchant settings, including client credentials, tokenization preferences, Google Pay and Apple Pay configurations, and fraud detection settings.

Default locale settings are managed separately through Site Preferences.

| + Create / Edit Configuration Refresh All Cache | | 2 Configuration(s) found |
|---|-------------|---|
| Locale | Merchant ID | Actions |
| en_CA | ██████████ | Edit Configuration Delete Configuration |
| en_US | ██████████ | Edit Configuration Delete Configuration |

Step 4: Fill in the Configuration Details

Now, enter the specific credentials for the selected locale.

- *Key Principle: You only need to fill in fields that are different from your site's default settings. Any field you leave blank will automatically use the value from your main Site Preferences.*
- *Required Field: Merchant ID is the only mandatory field.*
- *Security Note: Sensitive fields will be masked (*****) after saving for security. If you don't change them, the existing saved value is preserved.*
- *Ensure you configure the custom site preference **`JpmcCheckoutMode`** to your local merchant site override configurations if you are managing mixed US/CA and EU sites within the same Business Manager instance.*

Step 5: Save and Verify Your Setup

- *Click Save Configuration. Your new settings are now active.*
- *To verify, place a test order on your storefront using the newly configured locale. Confirm the payment is processed successfully and check the order details in Business Manager to ensure the correct Merchant ID was used.*

12.4 Management

Part 2: Management

The **JPMC Tools** → **Multi-Merchant Configuration** page is your central hub for managing all setups.

| Action | How to Do It |
|------------------------|--|
| Edit a Configuration | Click Edit Configuration next to the locale you wish to update. Changes are live immediately after you save. |
| Temporarily Disable | Edit the configuration and set the Enabled toggle to No. The locale will temporarily use the site's default account. |
| Delete a Configuration | Click Delete Configuration. The locale will permanently revert to using the site's default account. |
| Force a Cache Update | Click Refresh All Cache on the main list page to ensure all storefront servers have the latest settings. |

12.5 Key Concepts

Part 3: Key Concepts Explained

How "Fallback" Simplifies Your Work

You do not need to fill in every single field for every locale. The system uses a smart "fallback" logic:

- *It first looks for a setting in your Locale-specific Configuration.*
- *If the field is blank there, it falls back to the value in your main Site Preferences.*
- *If it's blank there too, it uses the built-in system default.*

This saves time and reduces duplication. You only need to define what's different.

The Importance of Order-Level Tracking

When an order is placed, the system saves the Merchant ID that was used directly on the order itself. This is critical because it means changes to your configurations do not affect past orders. If you need to capture or refund an old order, the system will always use the correct credentials that were active when the order was first placed.

12.6 Troubleshooting Multi-Merchant Issues

Part 4: Troubleshooting Common Issues

| If you see this problem... | ...try this solution: |
|--|---|
| <i>A locale is using the wrong account.</i> | Go to its configuration and ensure the Enabled toggle is set to Yes. Then, click the Refresh All Cache button on the main configuration page. |
| <i>The "Create / Edit" page shows "Disabled".</i> | The main feature is turned off. Follow the instructions in Part 1, Step 2 to enable it in Site Preferences. |

| | |
|--|---|
| <i>I deleted a configuration by mistake.</i> | Don't worry, this does not affect past orders. New orders on that locale will now use the site's default credentials. You can recreate the configuration at any time. |
| <i>The "Select Locale" dropdown is empty.</i> | Your site has no locales configured other than default. You must first add locales to your site in Merchant Tools → Site → Site Settings → Locales. |

13. PAYMENT METHODS SETUP

Prerequisite: This section assumes you have successfully completed the Site Import process outlined in the Metadata Import section. The payment processors and methods must be imported.

Please verify the following configurations to ensure that checkout transactions are routed correctly to the J.P. Morgan gateway.

13.1 Ensure Credit Card is enabled in Payment Method

- *Merchant Tools > Ordering > Payment Methods*
- *Locate `CREDIT\_CARD` (Has to be enabled)*
- *Ensure Enabled*
- *Verify/Set Payment Processor to `JPMC\_Payment`*
- *Apply changes*

[Merchant Tools > Ordering > Payment Methods](#)

Payment Methods

| Payment Methods | | | |
|---|---------------|---------|------------|
| To learn about Salesforce Payments, see here . | | | |
| Payment methods are managed here. To create a new payment method, click the New button. To remove a payment method click the remove icon in the payment method row. The default payment methods can't be removed, and their IDs can't be changed. When you select the CREDIT_CARD payment method, credit/debit cards can be reordered through drag-and-drop. | | | |
| New Sort Order Credit/Debit Cards Import/Export | | | |
| ID | Name | Enabled | Sort Order |
| AdyenComponent | Adyen Payment | Yes | 7 |
| BANK_TRANSFER | Bank Transfer | No | 8 |
| BMC | Bill Me Later | No | 12 |
| CREDIT_CARD | Credit Card | Yes | 10 |

13.2 Verify Accepted Card Types

| Card Type | Card Brand |
|------------|------------------|
| Visa | Visa |
| Mastercard | Mastercard |
| Amex | American Express |

| | |
|-------------|-------------|
| Diners Club | Diners Club |
| Discover | Discover |
| JCB | JCB |

13.3 Ensure Apple Pay and Google Pay are enabled in Payment Method

- *Merchant Tools > Ordering > Payment Methods*
- *Locate `DW\_APPLE\_PAY` and JPMC_GOOGLE_PAY` (Has to be enabled)*
- *Ensure Enabled*
- *Verify/ Set Payment Processor to `JPMC\_Payment`*
- *Apply changes*

Payment Methods

Payment methods are managed here. To create a new payment method, click the **New** button. To remove a payment method click the remove icon in the payment method row. The default payment methods can't be removed, and their IDs can't be changed. When you select the CREDIT_CARD payment method, credit/debit cards can be reordered through drag-and-drop.

Language: Default

| ID | Name | Enabled | Sort Order |
|------------------|------------------|---------|------------|
| BANK_TRANSFER | Bank Transfer | No | 3 |
| BML | Bill Me Later | No | 7 |
| CREDIT_CARD | Credit Card | Yes | 5 |
| DW_ANDROID_PAY | Android Pay | No | 2 |
| DW_APPLE_PAY | Apple Pay | Yes | 1 |
| GIFT_CERTIFICATE | Gift Certificate | No | 4 |
| JPMC_GOOGLE_PAY | Google Pay | Yes | 8 |

13.4 Configure Apple Pay in SFCC Business Manager

This configuration is mandatory for enabling Apple Pay within the SFRA storefront. Proper setup is required to ensure the Apple Pay payment sheet initializes and submits correctly.

- *Go to **Merchant Tools > Site Preferences > Apple Pay***
- *Configure the built-in Apple Pay settings:*
- *Required Details:*
 - *Ensure Apple Pay is Enabled? * - MUST*

- *Apple Merchant ID*: Enter the exact Merchant ID registered within your Apple Developer Portal.*
- *Apple Merchant Name*: Enter the display name of your store/brand as it should appear to the customer.*
- *Country Code*: Eg: US*
- *Merchant Capabilities*: You must select 3DS. (Note: Although marked as optional in interface, 3DS capability is strictly required for the Apple Pay sheet to submit successfully via this integration).*
- *Supported Networks*: Supported card Types*
- *Ensure Business Manager is registered with Apple pay. (Use `Register Sandbox` option). Also ensure, the domain registered with Apple and the site domain in which Apple pay is required must match. For any mismatches, add Site Aliases under Merchant Tools -> SEO & Discoverability*

13.4 Ensure 'Jpmc Drop In' is enabled in Payment Method (Only if Site /any locale needs to use Drop-in)

Merchant Tools > Ordering > Payment Methods

- *Locate `JPMC\_DROP\_IN` (Has to be enabled)*
- *Ensure Enabled*
- *Verify/ Set Payment Processor to `JPMC\_Payment`*
- *Apply changes*

14. FRAUD CHECK CONFIGURATION (OPTIONAL) – DIRECT API ONLY

14.1 Overview

The library integrates with Kount for device fingerprinting. When enabled:

1. Kount SDK loads automatically when `JPMCCheckoutProvider` mounts
2. Device data is collected under a session ID.
3. Session ID is sent in the Fraud Check request.
4. JPMC evaluates fraud risk and returns action (Accept/Review/Decline)

14.2 Enable Fraud Check

- **Merchant Tools > Site Preferences > Custom Preferences**
- *Open JPMC Payment Configuration*
- *Set Enable JPMC Fraud Check (Handle) = `true`*
- *Click Save*
- *Optionally Set Enable JPMC Fraud Check (Authorize) = `true`*

14.3 Multi-Merchant Configuration for Fraud Check

- *If your storefront operates in multi-merchant mode, you should enable Fraud Check flags independently for each merchant locale.*
- *Default Value: False*

14.4 Fraud Rule Action

| Action | Behavior |
|--------|----------------------------------|
| A | Accept. Proceed to authorization |
| E, R | Proceed; flag for manual review |

D

Decline before authorization

15. 3DS CONFIGURATION

15.1 Enable 3DS

- **Merchant Tools > Site Preferences > Custom Preferences**
- *Open JPMC 3D Secure Configuration*
- *Set Enable JPMC 3D Secure = `true`*
- *Click Save*

15.2 Multi-Merchant Configuration for 3DS

- *If your storefront operates in multi-merchant mode, you should enable 3DS flag independently for each merchant locale.*
- *Default Value: False*

****See the Multi Merchant section for reference.**

16. ORDER MANAGEMENT IN BUSINESS MANAGER (CSC)

16.1 Enable this feature

- Go to **Administration > Organization > Roles & Permissions > SELECT A ROLE (Eg: Administrator) > Customer Service Center Permissions**
- **Select Site (context)**
- **Click JPMC Payments**
- **Click Update**

16.2 Access Customer Service Center

- **Navigate to Merchant Tools > Ordering > Customer Service Center.**
- **Use the Find Order search functionality to locate and open the desired customer order.**
- **In the upper-right action menu, click on More... and select JPMC Payments from the dropdown list to access the transaction management view.**

| | | | | | |
|------------------------|---|--|--|---|-------------------------|
| Order No.:
00008201 | Creation Date:
3/3/2026 11:51 pm
Source Code: | Order Status:
Status: OPEN
Confirmation Status: NOTCONFIRMED
Export Status: NOTEXPORTED
Shipping Status: NOTSHIPPED
Payment Status: NOTPAID
Is A Gift: ☺ | Billing Address:
env vars test
env vars test
United States 77056
Houston TX
env-vars@test.com | Payment Method:
CREDIT_CARD Visa
Details... (1) | More... |
|------------------------|---|--|--|---|-------------------------|

| SHIPMENT 1 | | NAME | AVAILABILITY | QUANTITY | PRICE | TAX | TOTAL |
|---|--|--|----------------|----------|----------|--------------|--------------------------|
| Ship To:
env vars test
env vars test
United States 77056
Houston TX | | Robert Ludlum's: The Bourne Conspiracy (for X-Box 360)
<small>sierra-the-bourne-conspiracy-xbox360M</small> | Available (89) | 1 | \$ 59.99 | \$ 3.00 | \$ 59.99 |
| | | | | | | | Shipping Method: Ground |
| | | | | | | Tax: \$ 0.30 | |

| | |
|--------------------------|--|
| Add Note | <div>ITEMS TOTAL: \$ 59.99</div> <div>SHIPPING TOTAL: \$ 5.99</div> <div>TAX TOTAL: \$ 0.30</div> <div>ORDER TOTAL: \$ 69.28</div> |
|--------------------------|--|

16.3 Available Actions

| Action | Description | When Available |
|---------|--|--------------------------------------|
| Capture | Capture authorized funds (full amount) | After authorization (MANUAL/DELAYED) |
| Refund | Refund captured payment (full/partial) | After capture |
| Void | Cancel authorization before capture | After authorization, before capture |

Note: If capture method is `NOW`, capture occurs at authorization and Capture is not shown.

JPMC Payments

JPMC Payment Management

| | | | |
|----------------|--------------------------------------|----------------------|--------------------------|
| Order Number | 00005401 | Payment Method | Credit Card |
| Order Total | CAD 866.72 | Payment Status | Authorized |
| Transaction ID | a5e3bc83-020e-4006-8290-c90cf80fd7af | | |
| Capture Method | MANUAL | Auth Timestamp | 2026-04-15T10:56:49.778Z |
| Authorized | Available for Capture | Available for Refund | |
| CAD 866.72 | CAD 866.72 | CAD 0.00 | |

Payment Actions

Capture

Capture Amount:

☐ Capture Full Amount

☐ Mark as final capture (no more captures after this)

Capture

Void Authorization

Amount to void: CAD 866.72

Void Authorization

Close

16.3.1 Capture a Payment

When to use:

If your capture method is `MANUAL`, the payment was only **authorized** (funds reserved) at checkout. You now need to **capture** (charge) the customer — typically when you ship the product.

- *Open the order in Business Manager.*
- *Navigate to the ****JPMC Payments**** section.*
- *You'll see the ****authorized amount**** and the ****available for capture**** amount.*
- *Enter the amount to capture (the full amount, or a partial amount for partial shipments).*
- *Click ****Capture****.*
- *The payment status updates. You can capture multiple times until the full authorized amount is used.*

| Payment Status | What It Means |
|----------------------------------|---|
| **A** (Authorized) | Payment approved, funds reserved, nothing charged yet. |
| **AC** (Auth & Captured) | Payment was charged immediately at checkout (using `NOW` mode). |
| **C** (Captured) | Full authorized amount has been captured. |
| **PC** (Partial Captured) | Some of the authorized amount has been captured; more remains. |

16.3.2 Issue a Refund

- *When to use: A customer wants their money back — either for a return, a complaint, or an error.*
- *Open the order.*
- *In the JPMC Payments section, you'll see the captured amount and the available for refund amount.*
- *Enter the amount to refund (full or partial).*
- *Click ****Refund****.*

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- *The refund is processed through J.P. Morgan, and the money is returned to the customer's card.*

| <i>Payment Status</i> | <i>What It Means</i> |
|--|--|
| **RF** (<i>Refunded</i>) | The full captured amount has been refunded. |
| **PRF** (<i>Partial Refunded</i>) | Some of the captured amount has been refunded; more remains. |

16.3.3 Void a Transaction

When to use: You want to cancel an authorized payment before any funds have been captured (e.g., the customer calls to cancel before you ship).

- *Open the order.*
- *In the JPMC Payments section, click Void*
- *The authorization is cancelled. No funds are captured.*

****Important:**** You can only void a payment that has ****not yet been captured****. Once funds are captured, you must issue a refund instead.

16.4 Transaction History

Every action (capture, refund, void) is logged in the order's JPMC transaction history. You can view:

- *Transaction ID*
- *Amount*
- *Timestamp*
- *Status*
- *Who performed the action*

This provides a complete audit trail for your records.

17. JPMC ACCOUNT UPDATER

17.1 Overview

The JPMC Account Updater service ensures that your customers' saved payment tokens remain valid even if their physical cards are re-issued or expired.

- *For Direct API (US/CA) sites, this card refresh is completed in real-time when a customer accesses their 'My Account' page.*
- *For Drop-in (EU) sites, card updates are handled asynchronously via platform notifications during active checkout transactions, as standard 'My Account' token storage is not supported in the Drop-in architecture.*

The JPMC integration supports two flows:

| Flow Type | Description |
|--|--|
| Asynchronous Webhook (Notifications) – DROP IN | JPMC actively pushes card-update notifications to your storefront. These updates are stored in a secure queue and processed automatically by a background scheduled job. |
| Real-Time Account Updater (RTAU) – DIRECT API | When a customer attempts to check out with a saved card, the JPMC gateway checks for updates in real-time. If new card data is available, it is returned in line with the payment response and applied to the order immediately. |

17.2 Supported Operating Modes (ONLY FOR DIRECT API)

You can control how the feature operates using the Account Updater Mode setting. This can be configured globally at the site level or customized on a per-merchant basis.

| Mode | System Behavior |
|-----------|---|
| NONE | The Account Updater feature is completely disabled (Default). |
| REAL_TIME | Only the Real-Time update during payment authorization is active. |

Fallback Logic: In a multi-merchant environment, the per-merchant configuration takes highest priority. If a merchant-specific setting is left blank, the system falls back to the global site preference. If neither is configured, the system defaults to NONE.

Note: The operating configurations `REAL\_TIME` rely on real-time API polling, which is only supported under the Direct API (PIE) architecture. For EU storefronts operating on Drop-in, the updates happen via job processing notifications.

17.3 Site Preferences Configuration for Account Updater (ONLY FOR DIRECT API)

To configure the global settings for your storefront, navigate to Merchant Tools → Site Preferences → Custom Preferences → JPMC Account Updater.

| Preference Name | Internal ID | Description |
|----------------------|-------------------------|-------------------------------------|
| Account Updater Mode | jpmcAccountUpdater Mode | Select REAL_TIME. Defaults to NONE. |

| | | |
|--|--|--|
| | | |
|--|--|--|

17.5 Multi-Merchant Configuration for Account Updater (FOR DIRECT API ONLY)

If your storefront operates in multi-merchant mode, you can configure Account Updater settings independently for each merchant locale.

****See the Multi Merchant section for reference.**

Within the Account Updater section of the record, you can configure the following overrides:

| Attribute | Description |
|----------------------|---|
| Account Updater Mode | Select NONE, REAL_TIME. If left blank, the site-level preference is used. |

Real-Time Flow (RTAU)

- Customer attempts to check out using an existing saved card.
- Storefront sends the standard authorization request to JPMC.
- JPMC processes the authorization and includes updated card data in line with the response (if an update exists).
- Storefront immediately applies the new token and/or expiry date to the saved card profile, and the checkout proceeds without interruption.

17.6 JOBS only needed for Dropin API

To process out-of-band payment states and order status updates securely, the Drop-in UI integration relies on a decoupled, two-stage background job mechanism. This architecture prevents transaction

processing logic from blocking the critical storefront checkout thread, ensuring storefront stability and reliability.

JPMC Notifications Poller (JPMCNotificationsPoll)

This job runs on a recurring schedule (recommended interval: every 5 minutes) to poll payment and checkout notification updates from the J.P. Morgan gateway. Rather than handling heavy database manipulation or triggering order processes inline, this step acts strictly as an ingestion layer. The retrieved transaction updates are parsed and staged safely in the custom object queue named

JPMCDropInOrdersNotifications.

JPMC Notification Queue Processor (JPMCNotificationQueueProcessor)

A separate, asynchronous worker job—scheduled to run concurrently at a matching interval (e.g., every 5 minutes)—monitors the **JPMCDropInOrdersNotifications** custom object. It sequentially processes each staged notification entry, updates the relevant order update, Customer Profile Id update, or payment updates inside Salesforce Commerce Cloud based on the gateway's instruction, handles post-processing, and logs confirmation results before removing or flagging the completed queues.

This ensures consistent error isolation and automatic retry capabilities should any transient network errors occur.

18. TESTING AND VALIDATION

18.1 Test Card Payments

Use J.P. Morgan sandbox test cards per J.P. Morgan developer [documentation](#).

18.2 Test Scenarios

| Scenario | Expected Result |
|--------------------------|----------------------------------|
| Successful authorization | Order placed; payment authorized |
| Capture (MANUAL) | Funds captured via CSC |
| Refund after capture | Refund processed via CSC |
| Void before capture | Authorization voided via CSC |
| Declined card | Order not placed; error shown |
| Fraud check — ACCEPT | Proceeds normally |
| Fraud check — REVIEW | Proceeds; flagged in logs |
| Fraud check — BLOCK | Declined pre-authorization |

18.3 Verify Service Communication Logs

- *Administration > Operations > Services*
- *Select a JPMC service*
- *Confirm Communication Log Enabled = `true`*
- *Administration > Site Development > Development Setup > Log Files*
- *Review logs prefixed `service-jpmc-`*

19. GO-LIVE CHECKLIST

19.1 Credentials and Configuration

- *Obtain production credentials (Client ID, Merchant ID, etc)*
- *Obtain production PIE ID*
- *Upload production certificate and private key*
- *Generate production `kid` thumbprint*
- *Update site preferences with production values*

19.2 Services

- *Update credential URLs from sandbox to production endpoints*
- *Administration > Operations > Services > Credentials tab*
- *Update each JPMC credential URL per J.P. Morgan production details*

19.3 Payment Methods

- *Confirm `CREDIT\_CARD` enabled with `JPMC\_Payment` processor*
- *Confirm accepted card types align to target market*
- *If using Google Pay: Is `JPMC\_GOOGLE\_PAY` payment method enabled with `JPMC\_Payment` processor*
 - *If using Apple Pay: Is Apple Pay enabled in both the built-in SFCC settings, and the payment method enabled with `JPMC\_Payment` processor*

19.4 Business Manager Paths

- *Storefront cartridge path:*
``int_jpmc_sfra:int_jpmc_core:app_storefront_base``
- *BM cartridge path:*
``bm_jpmc:int_jpmc_core:bm_app_storefront_base``

19.5 Testing

- *Complete production end-to-end payment for all payment Methods*
- *Validate capture/refund/void in CSC*
- *Validate fraud thresholds (if enabled)*
- *Review communication logs*

19.6 Security

- *Confirm PIE encryption (no plaintext card data)*
- *Confirm CSRF protection on checkout forms*

19.7 Multi-Merchant (if applicable)

- *Is the ****Enable Multi-Merchant**** preference turned on?*
- *Are locale-specific configurations created and ****enabled**** for each required locale?*

20. TROUBLESHOOTING

20.1 Authorization Fails

| Possible Cause | Resolution |
|------------------------------------|--|
| Invalid/expired OAuth2 credentials | Validate Client ID, Resource ID, etc in site preferences |
| Certificate mismatch | Re-upload cert/key; regenerate `kid` |
| Incorrect service URLs | Verify in **Services > Credentials** |
| Service disabled | Ensure JPMC services are enabled |

20.2 PIE Encryption Not Working

| Possible Cause | Resolution |
|-------------------------------|---|
| Incorrect PIE URLs and PE key | Validate Get Key URL, Encryption URL and PE Key in site preferences |
| JavaScript errors | Check browser console for PIE errors |

20.3 Capture/Refund/Void Fails in CSC

| Possible Cause | Resolution |
|-------------------------------------|---|
| Missing BM cartridge path | Confirm
`bm_jpmc:int_jpmc_core` is present |
| Transaction already captured/voided | Check payment transaction state |

20.4 Viewing Logs

| Log Type | Business Manager Location |
|----------------------------|---|
| Custom error logs | Log Files: `customerror-*` |
| Service communication logs | Log Files: `service-jpmc-*` |
| Custom info/warn logs | Log Files: `custominfo-*` /
`customwarn-*` |

20.5 Regional Checkout Failures/ Payment UI Mismatch

| Symptom / Error | Root Cause Analysis | Corrective Action Steps |
|---|---|--|
| <ul style="list-style-type: none"> Customers in the EU see a broken Direct API form instead of Drop-in. US/CA checkout fails to render credit card forms. Error logs show "Invalid integration model requested." | <p>The site-level preference `JpmcCheckoutMode` is misconfigured for the current site locale (e.g., PIE has been assigned to an EU site).</p> | <ol style="list-style-type: none"> 1. In Business Manager, navigate to Merchant Tools > Site Preferences > Custom Preferences > JPMC Payment Configuration. 2. Locate the attribute JPMC Checkout Mode (`JpmcCheckoutMode`). 3. Verify that the value is set to PIE for US/CA sites, and DropIn for EU sites. 4. If utilizing a Multi-Merchant configuration, verify that local custom overrides are not forcing an unsupported integration type for the target locale. |

21. SUPPORT AND ESCALATION

- *J.P. Morgan API issues: contact your J.P. Morgan technical account representative.*
- *Cartridge installation / SFCC platform issues: contact your SFCC system integrator.*

APPENDICES

APPENDIX A. SAMPLE 'DW. JSON'

```
{  
  "hostname": "your-sandbox.demandware.net",  
  "username": "your-bm-username",  
  "password": "your-bm-password",  
  "code-version": "version1"  
}
```

APPENDIX B. BUILD AND UPLOAD COMMANDS

```
```bash
```

npm install
-------------

npm run build
---------------

```
```
```

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